M.Sc. thesis projects on NLP and related topics

Deep Natural Language Processing
Prof. Luca Cagliero
A.A. 2025/2026

Turin, 3/12/2025

Politecnico di Torino

Our group

- **Context**
  - We are members of **Database and Data Mining** (**DBDM**) **group** led by prof. Elena Baralis
  - The research activities of our group are mainly in the areas of Data Science, Machine Learning, and Big Data
  - This is a non-exhaustive list of currently available M.Sc. projects related to NLP
- **How to contact us**
  - Institutional email (name.surname@polito.it)

NLP, Multimodal Learning, and Related Areas

- **Luca Cagliero**, Associate Professor
- **Lorenzo Vaiani**, PostDoc
  - Multimodal Learning
- **Giuseppe Gallipoli**, PhD
  - Summarization, Conversational AI
- **Davide Napolitano**, PhD
  - Multimodal Learning
  - Neural Shapley Estimation
- **Simone Papicchio**, Joint PhD with EURECOM
  - co-supervised by **Prof. Paolo Papotti**
  - Text2SQL
  - LLMs for Tabular Data Understanding
- **Aurora Gensale**, PhD
  - Agentic AI
  - AI for Cybersecurity
- **Ali Yassine**, PhD
  - Time Series Anomaly Detection
  - Predictive Maintenance
- **Andrea Cacioli**, PhD
  - Legal AI
  - Agentic AI








Industrial Collaborations

- **Tierra**
  - Anomaly Detection from time series, Predictive Maintenance
- **FiberCop**
  - LLMs and RAG systems
- **Intesa Sanpaolo**
  - AI for Anti-Financial Crime
  - Visually-Rich Document Question Answering
  - Document summarization for Trend and Risk Analysis
- **DriveSec**
  - AI for CyberSecurity
- **ASL Torino 1**
  - Legal AI
- **Jakala**
  - Agentic AI
  - RAG systems

Research topics outline (1/2)

- Temporal reasoning
  - (Multimodal) LLMs and Transformers applied to text, video, speech, images
- Summarization
  - (Multimodal) LLMs and Transformers applied to text, video, and speech data
- Visually rich document understanding
  - Visual LLMs and Visual Transformers applied to documents including tables, figures, and other layout elements
- Fake news and hate speech detection
  - Visual LLMs and Visual Transformers used to process pairs of images and text
- Anomaly detection from time series
  - Time Series LLMs and Transformers to analyze time series
- Figurative Language Understanding and Explanation
  - LLMs applied to texts including non-literal content requiring deeper understanding

Research topics outline (2/2)

- Neural Shapley Estimation with LLMs
  - Neural XAI methods and LLMs to process images and tabular data
- Text2SQL
  - LLMs and Agentic AI to process tabular data
- Human-AI Agentic Collaborative Scientific Program Committee
  - Agentic AI for document processing
- AI models to accurately measure the human boby
  - Visual and Video LLMs to analyze images and videos

Temporal reasoning with (Multimodal) LLMs

- **Assess** the capabilities of state-of-the-art textual, Visual, and Speech LLMs in understanding temporal relations and complex time-related concepts
- Release **new benchmarks** for temporal reasoning
- Define appropriate **mitigation strategies**
    - In-context Learning
    - Knowledge Injection
    - Supervised Fine-Tuning
    - Reinforcement Learning techniques (e.g., DPO)
- Develop and release **new** lightweight **models** specialized for temporal reasoning
- Define **new evaluation metrics** for temporal reasoning model assessment

Temporal reasoning with (Multimodal) LLMs

- **Test** the capabilities of state-of-the-art textual, Visual, and Speech LLMs in understanding temporal relations and time-related concepts
- Release **new benchmarks** for temporal reasoning
- Define appropriate **mitigation strategies**

Project recently funded by the Italian Ministry:

LANGUAGE MODELS: A MATTER OF TIME

Fondo Italiano per la Scienza
https://fis-submission.mur.gov.it/bando-fis-2/

Several open PhD positions available for the upcoming 2026 call
https://www.polito.it/en/education/phd-programmes-and-postgraduate-school/admissions-to-phd-programmes/admissions/call-for-applications

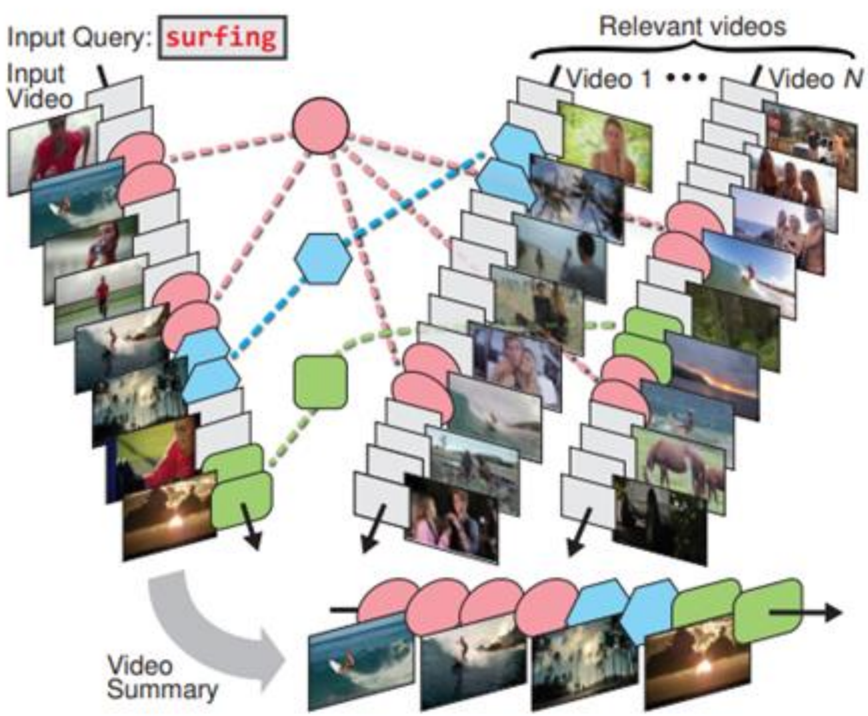
- D… poral re…
- Define **new evaluation metrics** for temporal reasoning model assessment

Video summarization

- **Benchmark public data**
  - (ours) https://aclanthology.org/2025.findings-acl.16/
  - https://github.com/yalesong/tvsum
  - https://epic-kitchens.github.io/2024
  - https://paperswithcode.com/dataset/summe
  - ...
- **Tasks**
  - **Challenge #1: Classical video summarization**: condense the key information conveyed by a video in a concise summary using VLLMs / Transformers

Video Summarization with LLMs

- **Challenge #3: Co-summarization with RAGs / Agentic AI**
  - Retrieve video moments relevant to several videos
    - E.g., https://openaccess.thecvf.com/content_cvpr_2015/papers/Chu_Video_Co-Summarization_Video_2015_CVPR_paper.pdf



Cross-lingual cross-temporal summarization

- **Challenge #1: handle temporal relations in news streams**
  - https://dl.acm.org/doi/10.1145/3158671
- **Challenge #2: Handle cross-lingual sources**
  - E.g., https://aclanthology.org/2024.cl-3.5/
- **Challenge #3: Handle multimodal sources**
  - https://doi.org/10.1145/3713070

Speech summarization

- **Benchmark public data**
  - Mega-SSum and CSJ-Ssum: https://arxiv.org/html/2408.00205v1
  - AudioSumm: https://www.kaggle.com/datasets/nfedorov/audio-summarization
  - ICSI: https://groups.inf.ed.ac.uk/ami/icsi/
  - AMI: https://groups.inf.ed.ac.uk/ami/corpus/
- **Tasks**
  - **Challenge:** speech summarization using paralinguistic and extralinguistic information

Opportunities of collaboration with UPC (Barcelona)

Visually Rich Document Understanding

- **VRDs**
  - Long documents integrating various types of visually rich elements such as paragraphs, tables, charts, diagrams, and photos
  - Common format: PDF files (natively digital or scanned)
- **Examples**
  - https://www.ijcai.org/proceedings/2024/690
  - https://github.com/adlnlp/pdfvqa

<u>Opportunities of collaboration with the University of Melbourne (Australia)</u>

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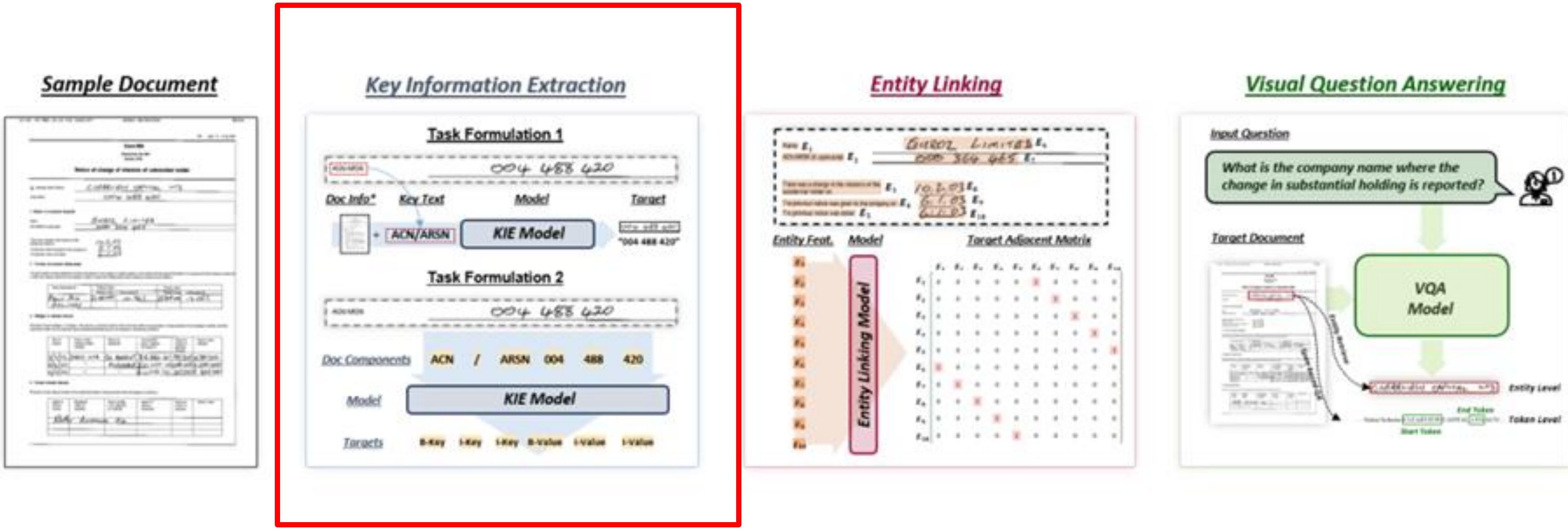
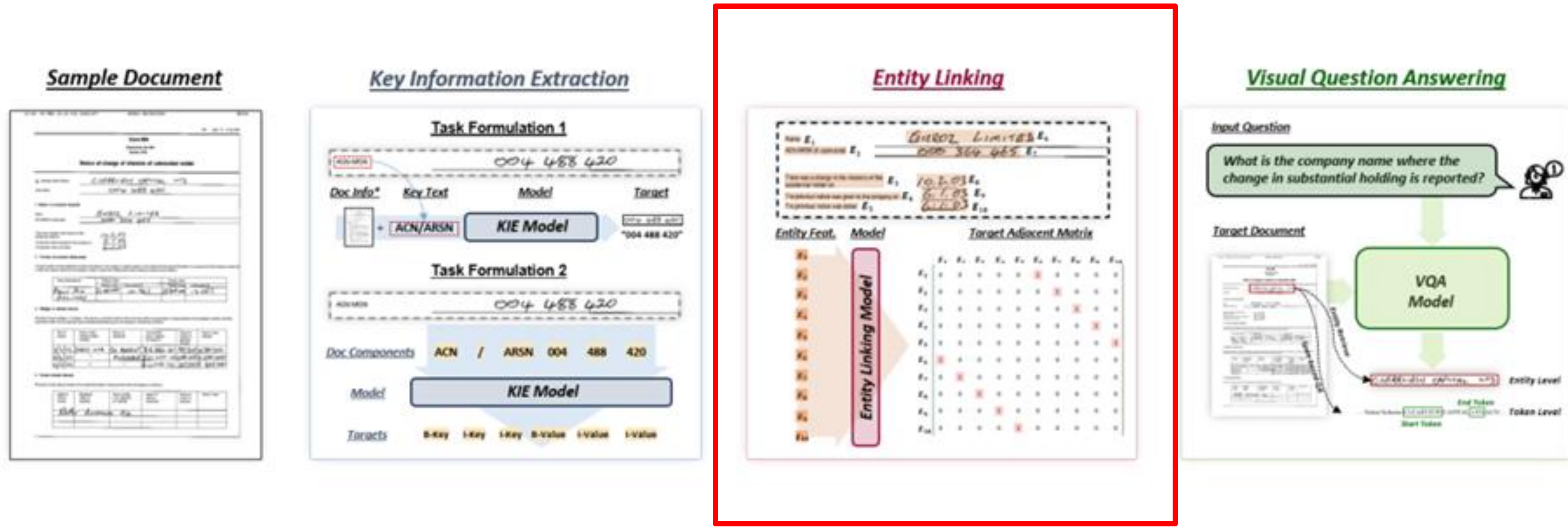
Visually Rich Document Understanding

- **Tasks**
  - **Key Information Extraction**: identify and extract values based on predefined keys



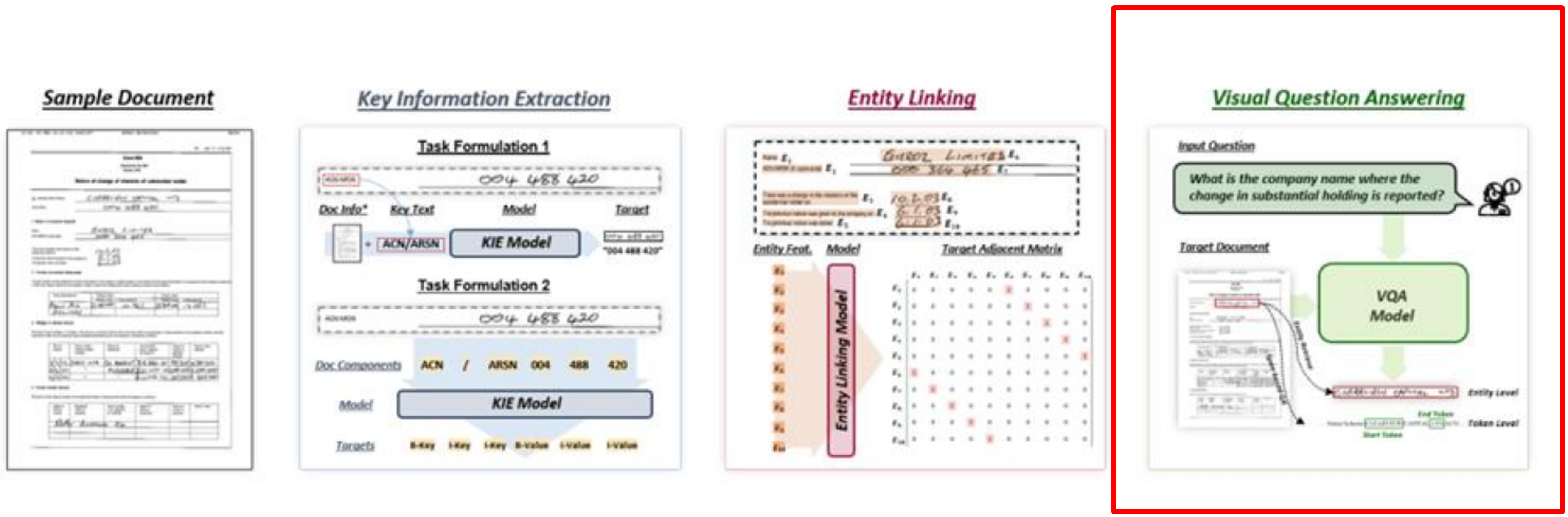
Visually Rich Document Understanding

- **Tasks**
  - **Entity Linking**: identify the semantic correlations between document entities (parent-child relations)



Visually Rich Document Understanding

- **Tasks**
    - **Visual Question Answering**: answering questions posed in natural language based on the contextual information in VRDs



Visually Rich Document Understanding

- **Challenges**
  - Detect and mitigate the main **weaknesses of VRD understanding models**
    - Unanswerable questions
    - Multi-modal entity relations
    - Multi-page entity relations
  - Design and testing of **ad hoc architectures** to analyze VRDs
    - Graph-based models, Teacher-Student models, etc.
      - E.g., https://arxiv.org/abs/2402.17983

Multi-Agentic Collaborative VQA Reasoning (1/2)

- **Concept**
  - Replace or augment a monolithic VLLM with a team of specialized agents, each focusing on a specific aspect of the reasoning process.
  - The agents work collaboratively to improve overall accuracy and robustness in identifying unanswerable questions.
- **Agents and Roles**
  - Layout Agent: Extracts and interprets structural relationships, element hierarchy, and reading order.
  - OCR/Text Agent: Handles textual understanding and entity linking.
  - Visual Agent: Processes figures, charts, and tables to extract quantitative or relational information.
  - Consistency Agent: Cross-checks outputs from all other agents and evaluates the answer's plausibility.

Multi-Agentic Collaborative VQA Reasoning (2/2)

- **Workflow**
  - Each agent independently analyzes the question-document pair.
  - Agents communicate through a structured dialogue (debate or consensus protocol).
  - The Consistency Agent consolidates evidence and provides the final answer and answerability score.
- **Benefits**
  - Modular reasoning that mirrors human reading strategies.
  - Improved interpretability, since each agent's contribution is explicit.
  - Enhanced robustness against specific corruption types (layout or visual).
- **Evaluation**
  - Compare multi-agent performance to single-model VLLMs.
  - Evaluate how agent communication protocols (e.g., ReAct, Tree-of-Thoughts) affect unanswerability detection.

Self-Reflective Agent for Answer Verification (1/2)

- **Concept**
  - This proposal focuses on training a unified model that internalizes the ability to reason about unanswerability. Instead of relying on an external verifier, the model learns during fine-tuning to detect missing or conflicting evidence within the document.
- **Workflow**
  - The model receives (Document, Question) and produces both an Answer and an Unanswerability Assessment.
  - The output includes an optional reasoning trace that explains why the answer is or is not supported.

Self-Reflective Agent for Answer Verification (2/2)

- **Benefits**
  - Internalizes evidence awareness: The model learns to reason about what information is missing.
  - Improved calibration: The model learns to express uncertainty explicitly.
  - Efficient inference: Single-pass output of both answer and verification.
- **Evaluation**
  - Compare with baseline VLLMs in terms of unanswerability detection accuracy.
  - Assess the quality and consistency of generated reasoning traces

Multi-Agent Reinforcement Learning for Document Exploration (1/2)

- **Concept**
  - Model document understanding as a multi-agent reinforcement learning (MARL) problem, where each agent explores specific sections or pages to gather evidence efficiently.
- **Agents and actions**
  - Page Agents: Each agent explores a subset of pages or regions.
  - Coordinator Agent: Aggregates findings and decides whether the question is answerable.

Multi-Agent Reinforcement Learning for Document Exploration (2/2)

- **Benefits**
  - Promotes efficient document traversal (less noise from irrelevant pages).
  - Scales to long multi-page documents.
  - Encourages cooperation between specialized agents.
- **Evaluation**
  - Compare exploration efficiency and answer accuracy with baseline models.
  - Assess the learned policies

Multimodal Fake News and Hate Speech Detection

- **Context**
  - proliferation of **fake news and hate speech** in online platforms
  - **Multimodal data sources** make the detection of fake news/hate speech content harder
    - fake/hateful contents can be hidden in either **visual** and **textual** news elements.
- **Challenge**
  - design and train a model capable of automatically recognize multimodal fake news from the joint analysis of **image** and **text**

Multimodal Fake News Detection

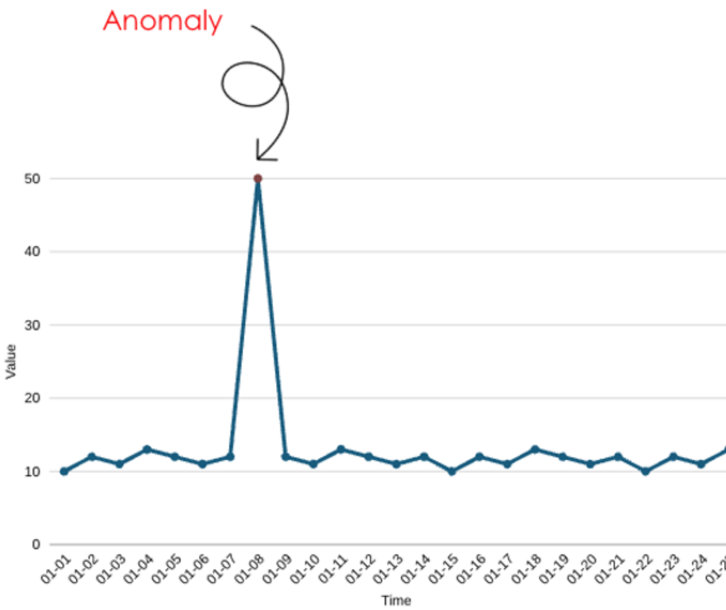
- **Surveys**
  - https://dl.acm.org/doi/pdf/10.1145/3700748
  - https://link.springer.com/article/10.1007/s13278-024-01280-3
- **Datasets**
  - Fakeddit, MediaEval, TI-CNN, Fauxtography
- **Challenges**
  - Benchmarking of Transformers and Multimodal LLMs
  - Analysis of the impact of external knowledge injection

Multimodal Hate Speech Detection

- **Recent work**
  - Images: https://arxiv.org/abs/2311.08110
  - Video: https://arxiv.org/abs/2508.20546
- **Datasets**
  - Meme Images
    - HatefulMeme, MMHS150k, MultiOff, MAMI
  - Meme Video
    - HateMM, ImpliHateVid
- **Challenges**
  - Benchmarking of Transformers and Multimodal LLMs
  - Analysis of the impact of external knowledge injection
  - Support textual LLM with multimodal content descriptions

AD-TS: Anomaly Detection in Time-Series Data

- What Is an **Anomaly**?
  A data point or pattern that does not match expected behavior (e.g., point or contextual)
- Why Does Anomaly Detection Matter?
  It **prevents failures** and **frauds**.
- Where It's Used? (EVERYWHERE)
  IoT sensors, finance, cybersecurity, healthcare, server monitoring
- Key Challenge
  Anomalies are **rare**, **diverse**, and highly **context-dependent**, making them hard to detect reliably.



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AD-TS: Anomaly Detection in Time-Series Data

- **Datasets**
  - Benchmark Time-Series Datasets
    - UCR (https://www.cs.ucr.edu/%7Eeamonn/time_series_data_2018)
    - mTDAS (https://github.com/2er0/mTADS)
    - TSB-UAD (https://github.com/TheDatumOrg/TSB-UAD)
  - **Financial** Time-Series Data
    - Stocks, currencies, ETFs, etc.
  - **Sensor / Industrial** Data
    - SynCAN (https://github.com/etas/SynCAN)

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AD-TS: Anomaly Detection in Time-Series Data

Challenges & Research Ideas

- **Benchmark Performance**
  - Evaluate and compare AD methods on various datasets
  - Adopt forecasting models for anomaly detection
  - Example models:
    - TimeLLM: https://arxiv.org/abs/2310.01728
    - Chronos-2: https://arxiv.org/pdf/2510.15821
- **Design Applications for Specific Use Cases**
  - Tailor anomaly detection to domains like finance, or vehicles
  - Explore practical pipelines like real-world deployment challenges
- **Combine LLMs with State-of-the-Art Methods**
  - Example 1: integrate LLMs with **Matrix Profile** techniques
  - Example 2: Use vLLMs to interpret Transformer-based models, providing explanations for detected anomalies

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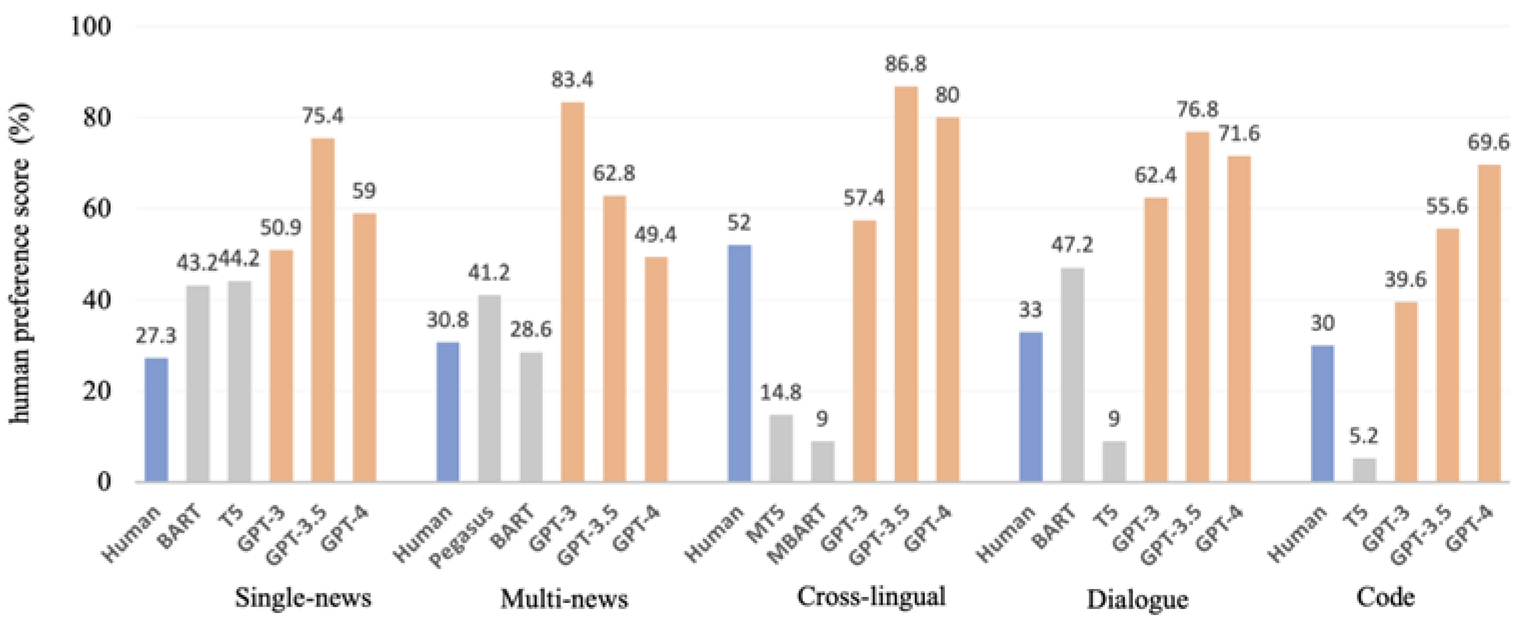
Text summarization with LLMs

- **Application of LLMs to**
  - **News summarization**: summarize a news article or an article collection
    - e.g., https://arxiv.org/abs/2309.09558
  - **Conversation summarization**: summarize email exchanges, threads of social posts or chats
    - e.g., https://arxiv.org/abs/2106.00829



Text summarization with LLMs

- **Challenge #1: Bias in Summarization**
  - Mitigate the presence of biased opinions, fake news, irrelevant aspects
- **Challenge #2: Input Order**
  - The sequential order of appearance of input tokens might influence summarization performance
- **Challenge #3: Retrieve & Summarize**
  - A preliminary retrieval stage may influence the result
  - Is retrieval (actually and always) beneficial for summarization? How does it affect factuality?
- **Challenge #4: Non-trivial Text Structure**
  - How to leverage the structure of the text to improve performance?
  - e.g., How can conversations be modeled more effectively?

Figurative Language Understanding and Explanation

- **Task**
  - Given a premise-hypothesis pair including figurative language, predict the type of relation (e.g., entailment or contradiction) and provide a textual explanation
    - e.g., FLUTE (https://aclanthology.org/2022.emnlp-main.481/)

Type	Premise (literal)	Hypothesis (figurative*)	Label	Explanation
Paraphrase + Sarcasm	My next door neighbors are *always arguing* in our shared hallway.	It's so *annoying* to have to hear my next door neighbors *argue all the time* in our shared hallway.	E	The sound of arguing neighbors can often be very disruptive and if it happens all the time in a common space like a shared hallway it is natural to find it annoying.
		It's so *pleasant* to have to hear my next door neighbors *argue all the time* in our shared hallway.	C	The sound of arguing neighbors can often be very disruptive and so someone considering it to be pleasant is not really accurate.
Simile	The assembly hall was now *hot and moist*, more so than usual.	In fact, the assembly hall was now *like a steam sauna*.	E	A sauna is a hot and moist environment, so the simile is saying that the hall is even hotter and more moist than usual.
	The assembly hall was now *cold and dry*, more so than usual.		C	A steam sauna is a small room or hut where people go to sweat in steam, so it would be hot and humid, not cold and dry.
Metaphor	He *mentally assimilated* the knowledge or beliefs of his tribe.	He *absorbed the knowledge* or beliefs of his tribe.	E	To absorb something is to take it in and make it part of yourself.
	He *utterly decimated* his tribe's most deeply held beliefs.		C	Absorbed typically means to take in or take up something, while "utterly decimated" means to destroy completely.
Idiom	Lady Southridge was wringing her hands, *trying hard and desperately to salvage* the bleak and miserable situation so that it somehow looks positive.	Lady southridge was wringing her hands, trying *to grasp at straws*.	E	To grasp at straws means to make a desperate attempt to salvage a bad situation, which is exactly what Lady Southridge is trying to do.
	Lady Southridge was wringing her hands, *doing absolutely nothing to overturn* the bleak and miserable situation so that it somehow looks positive.		C	To grasp at straws means to make a desperate attempt to salvage a bad situation, but the sentence describes not doing anything to change the situation

Figurative Language Understanding and Explanation

- **Challenge #1: Benchmarking**
  - Extend the dataset towards multilingual scenarios
  - Extend the analysis towards additional types of figurative language
- **Challenge #2: Explainability**

  Leverage model reasoning or state-of-the-art XAI tools to improve LLM performance
  - Parallel directions
    - Explain the prediction of the type of sentence relation
    - Explain the classification of figurative language types
    - "Explain the explanations" produced by the models
- **Challenge #3: Robustness**

  Test sensitivity to syntactic, lexical, or semantic perturbations in figurative expressions

Figurative Language Understanding meets... ...Text Summarization

- **Two established tasks**: certainly true for text summarization, but less so for figurative language understanding/explanation
- **Figurative language tasks** generally involve figurative expressions **in isolation**
  - e.g., figurative language classification, explanation, sarcasm detection, sentence-pair NLI, ...
- However, there is a lack of analyses involving figurative language within broader NLP tasks
  - What happens when we try to **summarize a text that contains figurative expressions**?
    - e.g., *The project meeting dragged on for hours, and by the end everyone felt the weight of the deadline <u>breathing down their necks</u>. Still, despite the pressure, the team managed to exchange creative ideas and agreed on a clear plan.*

Figurative Language Understanding meets… …Text Summarization

- **Challenge #1: Task definition & Dataset construction**
  - What do we aim to evaluate? What summarization requirements do we want to set?
  - Construct a dataset for the task (from scratch and/or by leveraging or combining existing datasets from the two areas)
- **Challenge #2: Benchmarking**
  - Analyze the performance of off-the-shelf (L)LMs on the task(s)
  - Evaluate multiple models and inference settings
- **Challenge #3: Development of tailored approaches**
  - Analyze the weaknesses of existing models & baseline techniques
  - Design ad hoc strategies to improve performance

Text Style Transfer

- **Task**: Transform the style of a piece of text while preserving its original content
- **Datasets**: Yelp, GYAFC, FlickrStyle10K, Shakespeare, ...
- **Challenges**: Extend existing benchmarks and adapt TST models to work in different scenarios
  - Standard binary setting: e.g., informal ↔ formal
  - Additional settings: Mixed-Style, Multi-Style, Cross-Lingual
  - Additional languages
- **Existing approaches**:
  - Survey: https://aclanthology.org/2022.cl-1.6/
  - Rule-based, disentanglement-based, prototype editing, (un)supervised, self-supervised (**ours**), reinforcement learning, ...

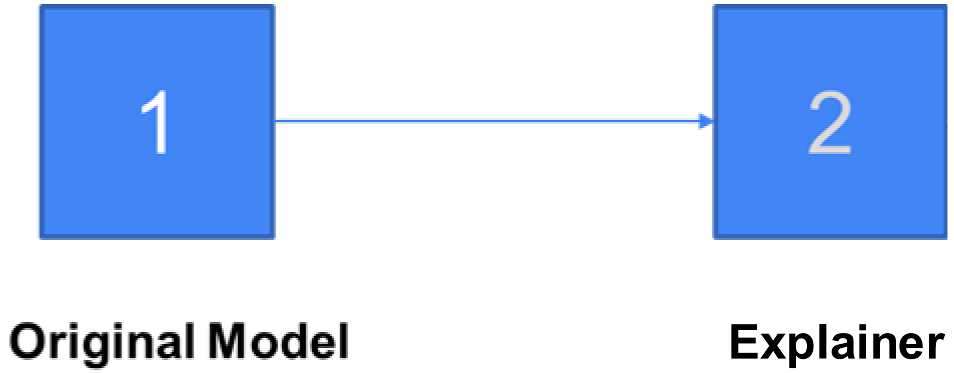
Text Style Transfer

- Extend or develop a **reinforcement learning**-based approach:
  - e.g., https://aclanthology.org/2025.naacl-long.135/
- Explore the integration of **DPO or GRPO/DAPO** techniques
  - Using **preference optimization** or **verifiable rewards**
  - For example, define **metrics or scoring functions** to provide the model with feedback without relying on annotated data
    - Using external models (e.g., style classifier)
    - Using proxy metrics (e.g., distance w.r.t. original sentence)
  - Integrating **model reasoning** in the process
- Which is more beneficial between human/AI feedback and automatic rewards? How to effectively reward (and penalize) the model?

Neural Shapley Estimation with LLMs

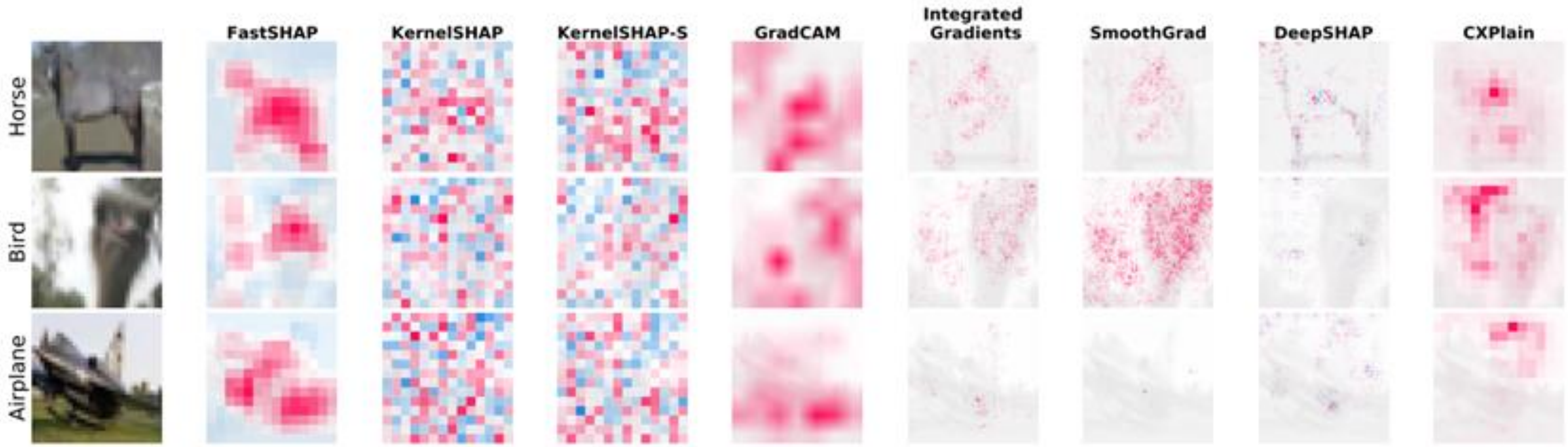
Shapley Values: Concept established for eXplainable AI

Used to calculate individual feature contribution to the model output



Neural Shapley Estimation with LLMs

- **Common datasets**
  - Tabular data
  - Images
  - Text
- **Examples**



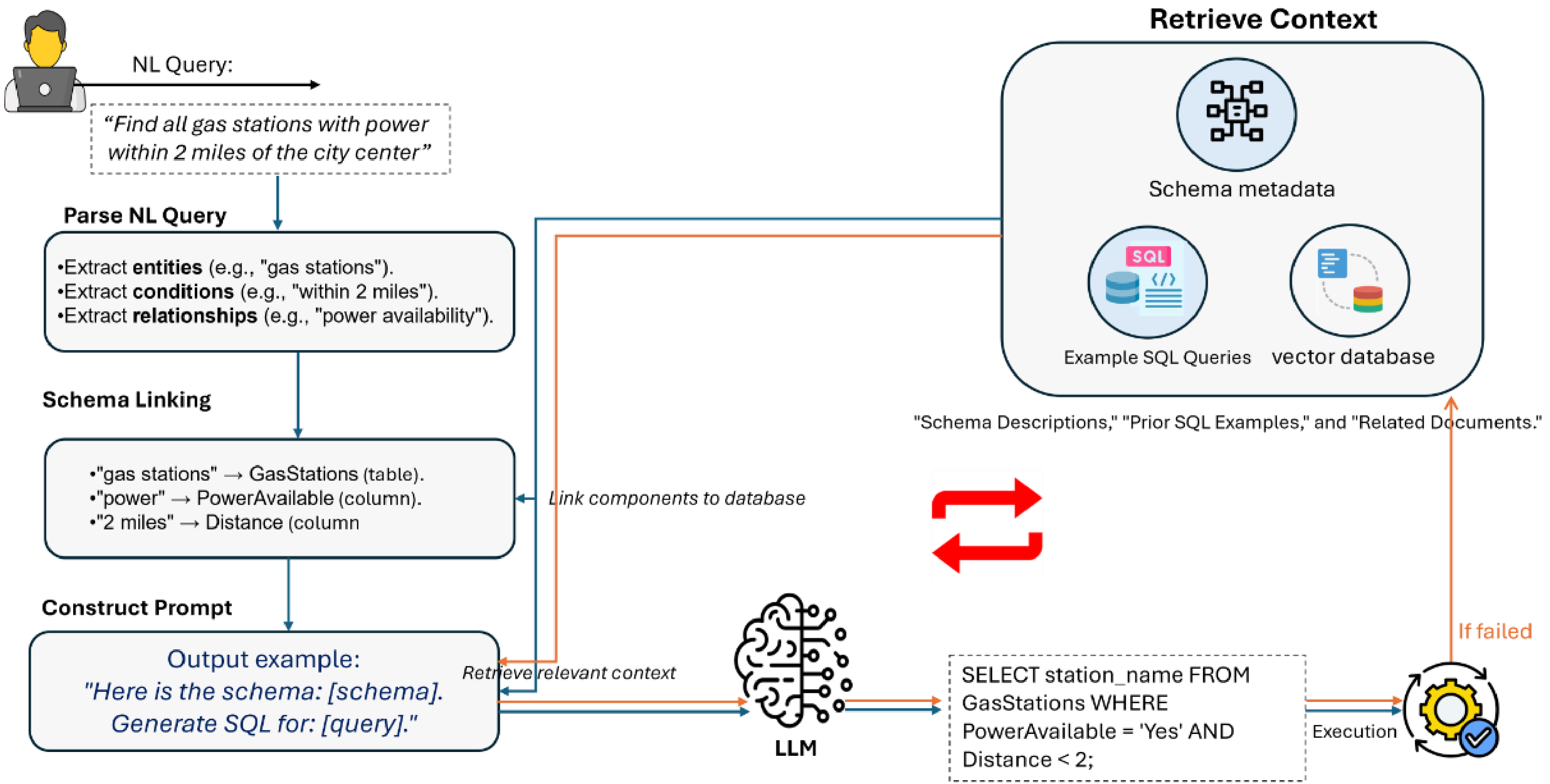
Neural Shapley Estimation with LLMs

- **Challenge:** Employ LLMs for Neural Shapley Estimation
  - Explain LLMs as black-box models
  - Use LLMs as self-explainer
  - Model Uncertainty between Modalities (Multimodal LLMs)

Text + Image → Multimodal LLM → Output

Is the Output aligned with both inputs?

Text2SQL: using LLMs to query databases



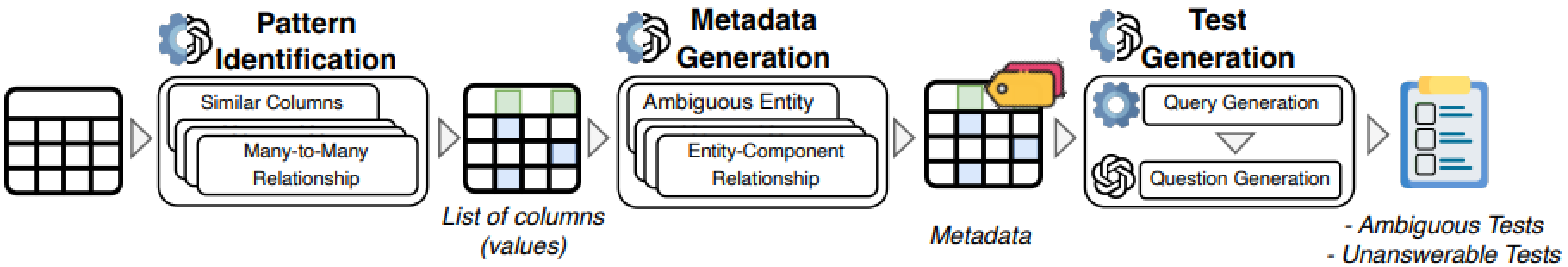
Text2SQL: using LLMs to query databases

- **Challenge #1: Text2Python**
  - When the natural queries are too complex SQL coding can be not enough...
- **Solution**
  - Python calls on relational DB (https://duckdb.org/)
  - Fine-tune Small LLMs using Reinforcement Learning

- **Challenge #2: Text2Cypher**
  - When the DB is not relational...
  - Query Graph DBs using LLMs
- **Approach**
  - Fine-tune Small LLMs using Reinforcement Learning

Opportunities of collaboration with Eurecom (France)

Evaluating LLM robustness to ambiguity in TextSQL



- **State of the art**
  - SQUAB https://aclanthology.org/2025.emnlp-main.906.pdf
- **Challenge #1**
  - Extend to NoSQL databases

Opportunities of collaboration with Eurecom (France)

Human-AI Agentic Collaborative Scientific Program Committee

- **Objective**
  - Design, develop, and test LLM-based **agentic AI solutions** to automate the process of scientific paper/slides revision
- Evaluation on **public benchmarks**
  - https://aclanthology.org/2023.acl-long.277.pdf
- Curation of **new datasets** based on
  - OpenReview (https://openreview.net/)
  - ACL Anthology (https://aclanthology.org/)

Opportunities of collaboration with the University of Darmstadt (Germany)

Development and Validation of AI model to accurately measure human body from images and videos

- Adopt GenAI models, including Visual and VideoLLMs, to take body measurements (e.g., arm and waist circumference)
- Main steps
  - Collection of training samples
  - Specialization of general-purpose models
  - Implementation and comparison with baseline Computer Vision approaches

Acknowledgements and copyright license

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- Affiliation
  - The author and his staff are currently members of the Database and Data Mining Group at Dipartimento di Automatica e Informatica (Politecnico di Torino) and of the SmartData interdepartmental centre
    - https://dbdmg.polito.it
    - https://smartdata.polito.it